📄 Software Requirements Specification (SRS)

AI Study Platform (StudyAI)

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1. Introduction

1.1 Purpose

This document provides a detailed description of the software requirements for the **AI Study Platform (StudyAI)**. It defines the system functionalities, constraints, and expected behavior. This document is intended for developers, testers, and stakeholders involved in the project.

1.2 Scope

StudyAI is a web-based application designed to assist students in learning using Artificial Intelligence. The platform allows users to upload study materials and leverage AI-powered features such as:

- Automatic summarization of notes
- Question generation
- AI-based chatbot for doubt resolution

The system aims to improve learning efficiency, productivity, and accessibility.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
UI	User Interface
DB	Database

1.4 Document Overview

This document describes the overall system, its functionalities, external interfaces, constraints, and non-functional requirements of StudyAI.

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2. Overall Description

2.1 Product Perspective

StudyAI is a standalone web application based on a **client-server architecture**:

- Frontend communicates with backend using REST APIs
- Backend handles business logic and AI processing
- Database stores user data and content

2.2 Product Functions

The system provides the following key functionalities:

- User registration and authentication
- Upload and management of study notes
- AI-powered summarization
- AI-based question generation
- Chat-based AI interaction
- Dashboard for tracking user activity

2.3 User Classes and Characteristics

User Type	Description
Student (Primary User)	Uses platform for study assistance, basic web knowledge required

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2.4 Operating Environment

- Web Browsers: Chrome, Edge, Firefox
- Operating Systems: Windows, Linux, macOS

2.5 Design and Implementation Constraints

- Requires stable internet connection
- Dependent on third-party AI APIs
- API usage may be limited based on plan

2.6 Assumptions and Dependencies

- Users have basic technical knowledge
- AI services remain available and responsive
- Database service is continuously accessible

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3. System Features (Functional Requirements)

3.1 User Authentication Module

- The system shall allow users to register using email and password
- The system shall allow secure login
- The system shall use JWT-based authentication

3.2 Notes Management Module

- The system shall allow users to upload notes (text or file)
- The system shall allow users to view their notes
- The system shall allow users to delete notes

3.3 AI Processing Module

- The system shall generate summaries of uploaded notes
- The system shall generate questions from notes
- The system shall provide a chatbot interface for user queries

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3.4 Dashboard Module

- The system shall display user activity

- The system shall show stored notes and AI interactions

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4. External Interface Requirements

4.1 User Interface

- Web-based interface accessible via browsers
- Clean, responsive, and user-friendly UI

4.2 API Interface

- RESTful APIs for frontend-backend communication
- JSON-based data exchange

4.3 Hardware Interface

- Standard computing devices (laptop/desktop)
- Internet connectivity required

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5. Non-Functional Requirements

5.1 Performance Requirements

- The system should respond within **2–3 seconds** under normal conditions

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5.2 Security Requirements

- User data must be protected
- Authentication must use JWT
- Sensitive data must be encrypted

5.3 Usability Requirements

- Interface should be user-friendly
- Navigation should be simple and intuitive

5.4 Scalability Requirements

- The system should support increasing number of users

5.5 Reliability Requirements

- The system should ensure minimal downtime
- High availability must be maintained

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6. System Architecture

The system follows a **layered architecture**:

- **Frontend:** React-based UI
- **Backend:** Node.js with Express
- **Database:** MongoDB
- **AI Integration:** External AI APIs

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7. Database Design

7.1 User Entity

- User ID
- Name
- Email
- Password
- Created Date

7.2 Notes Entity

- Note ID
- User ID
- Title
- Content
- Created Date

7.3 AI Interaction Entity

- Interaction ID
- User ID
- Query
- Response
- Timestamp

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8. Future Enhancements

- Mobile application version
- Voice-based AI interaction
- Personalized study recommendations
- Multi-language support